



SIMATS Online Programming Lab

JAVA PROGRAMMING



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QUESTIONS

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Pro-Consume

PRO-CONSUME

Write a Java program to create a producer-consumer scenario using multiple threads. The producer thread should add items to a queue, and the consumer thread should remove items from the queue and process them.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.concurrent.ArrayBlockingQueue;
2 import java.util.concurrent.BlockingQueue;
3
4 public class ProConsume {
5
6     public static void main(String[] args) {
7         // Create a shared queue with a capacity of 5
8         BlockingQueue<Integer> queue = new ArrayBlockingQueue<>(
9
10        // Producer Thread
11        Thread producer = new Thread(() -> {
12            try {
13                for (int i = 1; i <= 10; i++) {
14                    System.out.println("Producer producing: " +
15                        queue.put(i); // Adds item to queue, waits i
16                        Thread.sleep(500); // Simulate time taken to
17                }
18            } catch (InterruptedException e) {
19                Thread.currentThread().interrupt();
20            }
21        }):
```

OUTPUT

Error: Execution timeout (>5 seconds) – possible infinite loop

INPUT

Your INPUT goes here!

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